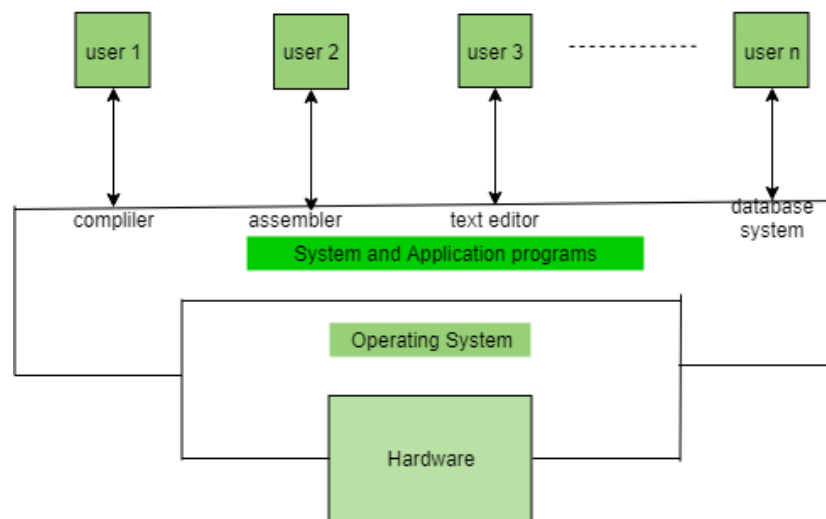


OPERATING SYSTEM

- An Operating System is System software that manages all the resources of the computing device. It acts as interface between the users and different parts of the computer or the computer hardware allowing you to interact with the machine without needing to understand its inner working.
- Manages the overall resources and operations of the computer.
- Controls and monitors the execution of all other programs that reside in the computer, which also includes application programs and other system software of the computer.

Definition: An operating system (OS) is software that manages computer hardware and software resources. It provides a stable environment for applications to run efficiently.



Operating System - Examples

There are plenty of Operating Systems available in the market which include paid and unpaid (Open Source). Following are the examples of the few most popular Operating Systems:

1. **Windows:** This is one of the most popular and commercial operating systems developed and marketed by Microsoft. It has different versions in the market like Windows 8, Windows 10 etc. and most of them are paid.
2. **Linux:** This is a Unix based and the most loved operating system first released on September 17, 1991 by Linus Torvalds. Today, it has 30+ variants available like Fedora, OpenSUSE, CentOS, Ubuntu etc. Most of them are available free of charges though you can have their enterprise versions by paying a nominal license fee.
3. **MacOS:** This is again a kind of Unix operating system developed and marketed by Apple Inc. since 2001.

4. **iOS:** This is a mobile operating system created and developed by Apple Inc. exclusively for its mobile devices like iPhone and iPad etc.
5. **Android:** This is a mobile Operating System based on a modified version of the Linux kernel and other open-source software, designed primarily for touch screen mobile devices such as smart phones and tablets.

FUNCTIONS OF OPERATING SYSTEM

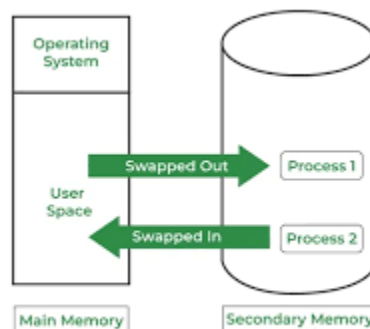
An Operating System (OS) is the backbone of a computer system. It acts as a manager for all resources and provides an interface between the user and the hardware. The OS performs several crucial functions to ensure smooth and efficient system operations.

1) Process Management

The OS manages all processes in the system. It handles process creation, scheduling, execution, and termination. It ensures fair CPU allocation among multiple processes.

2) Memory Management (Main Memory Management)

The OS keeps track of memory usage by various processes and allocates or deal locates memory as required. It ensures optimal utilization and prevents memory conflicts.



3) I/O Device Management

The OS manages input/output devices through drivers and buffers. It schedules and controls all I/O operations, ensuring smooth communication between hardware and software.

4) File Management

The OS organizes data into files and directories, controls file access permissions, and manage file storage and retrieval on different devices.

5) Secondary Storage Management

The OS supervises storage devices such as hard drives. It handles data storage, retrieval, and organization for long-term use.

6) Network Management

The OS facilitates data sharing and communication among computers in a network. It manages network connections, protocols, and data transmission securely.

7) Security Management

The OS ensures data and resource protection by managing user authentication, access control, and encryption mechanisms to prevent unauthorized use.

8) Command Interpreter System (User Interface)

The OS provides a way for users to interact with the system through Command-Line Interface (CLI) or Graphical User Interface (GUI).

9) Control over System Performance

The OS monitors system performance, manages resource usage, and ensures efficient task execution.

10) Job Accounting

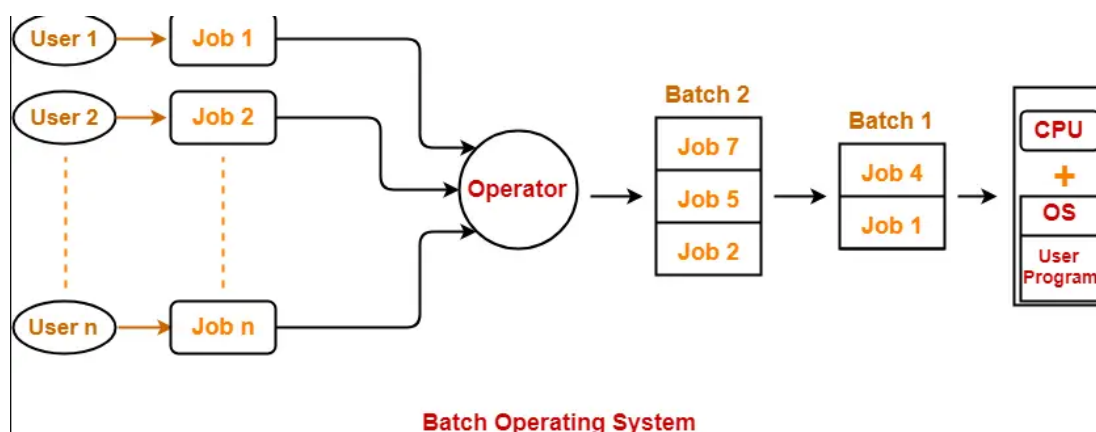
The OS keeps track of resource usage for various users and processes. This helps in system auditing and cost analysis.

11) Error Detection and Correction

The OS continuously checks for hardware and software errors. It takes corrective measures or reports errors to maintain system reliability.

12) Coordination between Other Software and Users

The OS ensures smooth coordination among users, applications, and hardware components, acting as a bridge between them.

TYPES OF OPERATING SYSTEMS**a) Batch Operating System (BPOS)**

- Firstly, user prepares his job using punch cards.
- Then they submit the job to the computer operator.
- Operator collects the job from different users and sorts the jobs into batches with similar needs.
Then operator submits the batches to the processor one by one.
- The system handles resource allocation for each job in the batch and releases the memory once the job is completed.
- All the jobs of one batch are executed together.

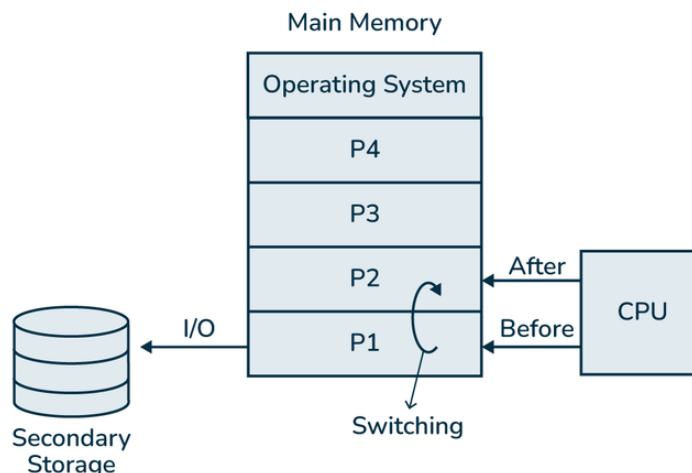
Advantages

- **Improved Throughput:** Batch systems can handle high volumes of jobs at once, improving overall system throughput.
- **Efficient Job Management:** Multiple users can efficiently share the system, making it cost-effective.
- **Minimal Idle Time:** The system minimizes idle time by processing jobs in a continuous sequence without human intervention.

Disadvantages

- **Inefficient CPU Utilization:** When a job is waiting for input/output (I/O), the CPU remains idle, leading to poor utilization of resources.
- **Increased Response Time:** The time between job submission and output can be high as all jobs are processed sequentially.

b) Multi-Programming Operating System



- A multiprogramming operating system which runs several programs on a single processor computer gives the CPU time to each program in turn so that a single processor computer can run more than one program.

- If a program has to wait for an I/O operation, other programs utilize the CPU in the meantime. These operating systems form an important and popular class of operating systems.
- Examples of multiprogramming operating system are Linux distributions, Windows, IOS, etc.
- In multiprogramming operating systems, while a program waits for some I/O operation to finish, some other program uses the CPU for that time being. It's, therefore, possible for many tasks to share the CPU time.

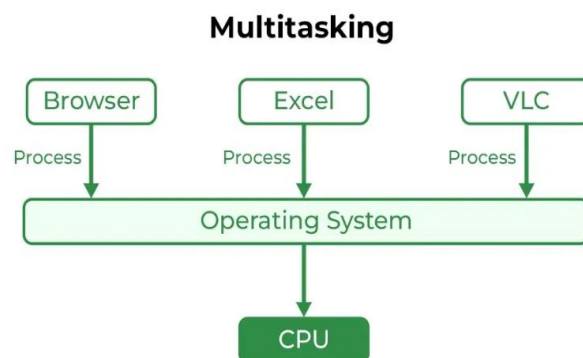
Advantages

- CPU is better utilized and the overall performance of the system improves.
- It helps in reducing the response time.
- Increase the throughput of the system.
- The waiting time for a program is reduced.
- Increase the resource utilization of a computer system.

Disadvantages

- **Increased complexity:** The OS must manage multiple programs, which requires sophisticated scheduling algorithms and memory management.
- **Resource contention:** Multiple programs competing for the same resources, such as memory, can lead to conflicts and performance degradation.

c) Multi-tasking/Time-sharing Operating systems



- Multi-tasking is a type of Multiprogramming system with every process running in round robin manner (a preemptive scheduling algorithm that allocates a fixed-size time slice, or "time quantum," to each process in a circular order).
- A multitasking operating system is capable of executing several programs simultaneously. The operating system achieves this by swapping each program in and out of memory, one at a time. When a program is brought out of memory, it is saved for a while on secondary memory until it is needed again.

- Each user gets the time of the CPU as they use a single system. These systems are also known as Multitasking Systems. The task can be from a single user or different users. The time that each task gets to execute is called quantum. After this time interval is over, the OS switches over to the next task.

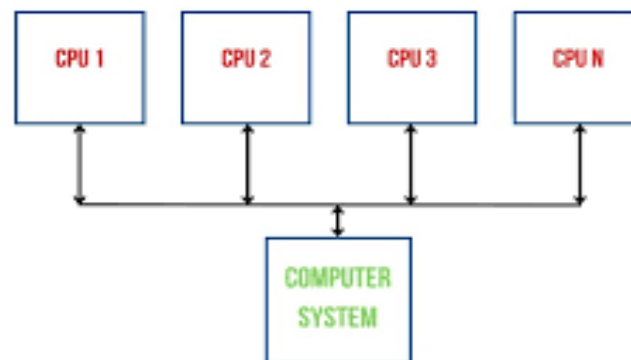
Advantages

- Each task gets an equal opportunity.
- CPU idle time can be reduced.
- Resource Sharing: Time-sharing systems allow multiple users to share hardware resources such as the CPU, memory and peripherals, reducing the cost of hardware and increasing efficiency.

Disadvantages

- Reliability problem
- Complexity: Time-sharing systems are complex and require advanced software to manage multiple users simultaneously. This complexity increases the chance of bugs and errors.

d) Multi-Processing Operating System

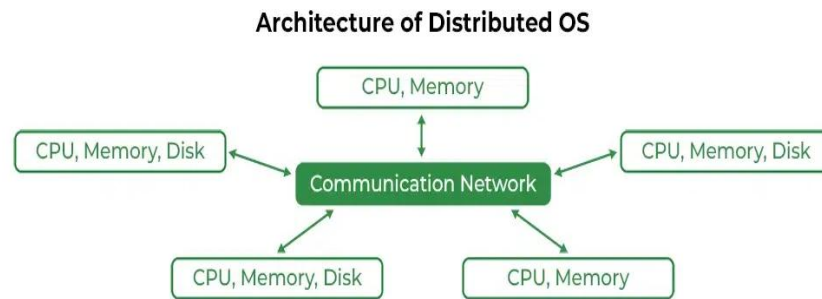


- Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It better the throughput of the System.
- Multiple processors work parallelly in multi-processing operating systems to perform the given task.
- All the available processors are connected to peripheral devices, computer buses, physical memory, and clocks.
- The main aim of the multi-processing operating system is to increase the speed of execution of the system.

Advantages

- It increases the throughput of the system as processes can be parallelized.
- As it has several processors, so, if one processor fails, we can proceed with another processor.

e) Distributed Operating System



- Various autonomous interconnected computers communicate with each other using a shared communication network.
- Independent systems possess their own memory unit and CPU. These systems processors differ in size and function.
- The major benefit of working with these types of operating systems is that it is always possible that one user can access the files or software which is not present on his system but on some other system connected within this network, i.e., remote access is enabled within the devices connected to that network.

Advantages

- Failure of one will not affect the other network communication, as all systems are independent of each other.
- Load on host computer reduces.
- Delay in data processing reduces.

Disadvantages

- Failure of the main network will stop the entire communication.
- These types of systems are not readily available as they are very expensive.

f) Real-Time Operating System

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**. **Real-time systems** are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Examples: Scientific experiments, medical imaging systems, Industrial control system, Weapon systems, Robots, Air traffic control systems

Types of Real-Time Operating Systems

- **Hard Real-Time Systems:** Hard Real-Time OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving lives like automatic parachutes or airbags which are required to be readily available in case of an accident. Virtual memory is rarely found in these systems.
- **Soft Real-Time Systems:** These OSs are for applications where time is less strict. Examples are multimedia streaming, online games, and digital cameras.

Advantages

- **Error-Free:** These types of systems are error-free.
- **Maximum Consumption:** Maximum utilization of devices and systems, thus more output from all the resources.

Disadvantages

- **Complex Algorithms:** The algorithms are very complex and difficult for the designer to write.

g) Mobile Operating System

A Mobile Operating System is one that has been developed for mobile devices such as mobile phones and tablets. These operating systems are touch-friendly, power efficient, and have software elements geared towards mobile devices, hence the convenience of the smaller handheld devices.

Examples: Android, iOS

Advantages:

- **User-friendly interface:** Optimized usage with the use of touch screen and motion.
- **Optimized for mobile hardware:** Power and resource consumption are well managed and properly utilized in order to conserve the available battery.
- **Wide range of apps:** Exploits the broad range of mobile applications available in the market.

Disadvantages:

- **Dependence on app stores:** Users are often limited to apps available in official app stores.

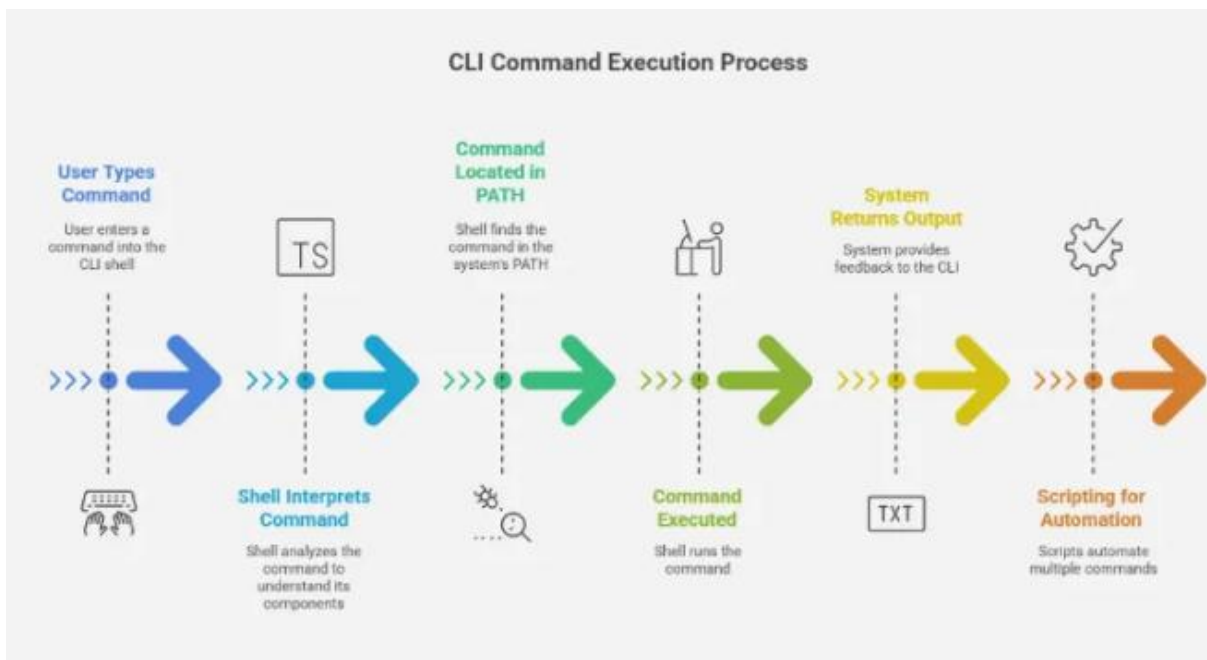
COMMAND BASED OPERATING SYSTEM OR COMMAND LINE INTERFACE

The Command Line Interface (CLI) is a text-based tool for interacting with a computer's operating system using typed commands. Unlike graphical interfaces (GUIs), CLI offers fast, precise control for managing systems and programs. *It is also known as character user interface.*

Examples

- **MS-DOS:** A classic example of a CUI operating system that was widely used on early personal computers.
- **UNIX:** A foundational operating system family whose command-line interface remains a key way to interact with its many variants.
- **Linux:** Many Linux distributions use a CUI in the form of the command line, which is essential for system administration and development.
- **Windows Command Prompt:** The built-in command-line interpreter for Windows operating systems is another prominent example of a CUI

It enables tasks like file management, system configuration, and automation without a graphical interface. Users type commands into a CLI shell (e.g., Bash, PowerShell), which interprets and executes user's commands. Shell acts as an intermediary between the user and the operating system. The shell parses the command, identifying the action, options, and arguments. It locates the command in the system's PATH and executes it. The system returns output (e.g., data, error messages) to the CLI.



MS-DOS Operating

MS-DOS Operating System also called the Microsoft Disk Operating system was developed by Microsoft for x86 personal computers. It is Command Line Interface (CLI) OS is a text-based OS.

Features

- **Command-line interface:** Users interact with MS-DOS by typing commands, such as DIR to list files or CD to change directories.
- **Disk-based:** It was named for its primary function of managing files and programs on a disk, loaded into memory from a floppy disk each time the computer booted.
- **Single-tasking:** Unlike modern operating systems, MS-DOS could only run one program at a time.

Common commands Used in MSDOS

Command	Description
DIR	Lists all files and folders in the current directory
CD	Changes the current working directory
DEL	Deletes one or more files
COPY	Copies files from one location to another
MOVE	Moves or renames files
REN	Renames files or directories
CLS	Clears the screen
FORMAT	Formats a disk for use

Graphical User Interface

GUI stands for graphical user interface. A GUI-based operating system uses graphical elements like icons, menus, and a mouse to allow users to interact with a computer, making it easier and more intuitive than text-based systems.

Examples

- **Microsoft Windows:** A family of operating systems for personal computers, well-known for its visual interface.
- **MacOS:** Apple's operating system, which has been a prominent example of a GUI since its introduction with the Macintosh.
- **Linux:** Many Linux distributions, such as Ubuntu, use a GUI as their primary interface, though they can also be run without one.
- **ChromeOS:** Google's operating system for Chrome books is built around a web-based graphical interface.

- **Mobile OS:** Many smart phone operating systems, like Android and Apple's iOS, are fundamentally GUI-based.

A graphical user interface (GUI) comprises visual components like cursors, icons, and buttons, enriched with sound and visual effects. These elements enable users to interact with a computer without needing to know specific commands. Below are the components of a Graphical User Interface:

- **Icons:** Small images representing files, folders, or actions.
- **Pointers:** Cursor symbols (e.g., arrows) for selecting items.
- **Pointing Tools:** Mouse or track pad to move pointers.
- **Menus:** Lists of clickable options for navigation.
- **Buttons:** Clickable elements to trigger actions.
- **Scrollbars:** Enables navigation through content beyond the visible area.
- **Window:** Contains applications or documents for user interaction.
- **Tab:** Separates content or functions within a single window.
- **Text Fields:** Input areas for typing data, such as search bars or form fields.
- **Toolbars:** Rows of buttons or icons for quick access to common commands (e.g., "Bold" or "Undo" in a text editor).

Components of a Graphical User Interface



Features

- **Visual interaction:** Users interact by clicking on graphical objects such as icons, buttons, and windows instead of typing commands.

- **Ease of use:** GUIs are generally easier to learn and use because they don't require users to memorize specific text commands.
- **Efficiency:** Shortcuts and visual cues can make tasks faster and more efficient.
- **Multitasking:** Most modern GUIs support multitasking, allowing users to work on multiple applications simultaneously.

Real-World Impact of GUIs

- **Healthcare:** Touch-based GUIs on medical devices make it easier and faster to enter patient data.
- **Gaming:** GUIs provide immersive experiences, enhancing game play on consoles and PCs.
- **Education:** Interactive GUIs in educational apps make learning fun and engaging for students.

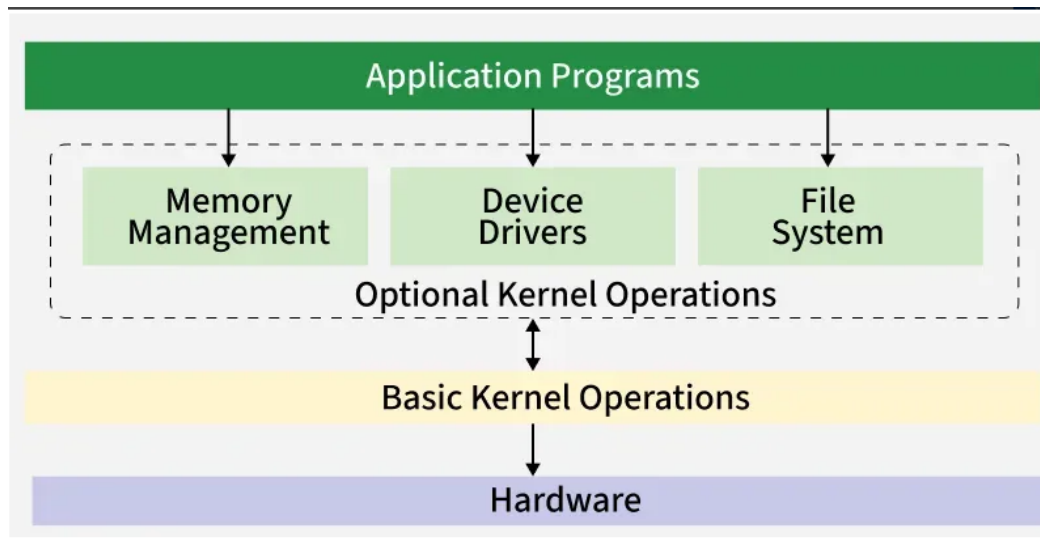
WINDOWS OPERATING SYSTEM

- Windows is an Operating System developed by Microsoft. First released in 1985.
- It is a graphical user interface (GUI) based system that allows users to interact with their computers and manage hardware and software.
- It runs applications and provides a user-friendly interface that enables users to interact directly with the system to perform tasks.

Features of windows operating system

- Users are allowed to run the software.
- Users interact with the OS through visual elements, such as icons, windows, menus, and a mouse-driven pointer.
- Users can view or store their files.
- Users can create, delete, rename, and move files using the GUI or the CLI.
- Windows OS versions are Windows 95, Windows 98, Windows NT, Windows XP, Windows Vista, Windows 7, Windows 8, Windows 8.1, Windows 10, Windows 11, Windows Server 2022

Architecture of Windows OS



- **Application Programs:** An Application Program is a type of computer software designed to perform specific tasks for users. Application Program helps users accomplish real-world activities such as creating documents, playing media, browsing the internet, or managing data
- **Memory Management:** Memory Management is a hardware component that stores data, instructions and information temporarily or permanently for processing. It consists of an array of bytes or words, each has a unique address. Memory management in the Windows Operating System is the process through which the OS handles, allocates, and frees physical and virtual memory to ensure efficient performance and multitasking.
- **Device drivers:** Device drivers are specialized software programs that act as a bridge between a computer's operating system and its hardware components.
- **File System:** A File System is defined as a medium used for saving and managing data in the computer system.
- **Kernel:** A kernel is the core part of an operating system. It acts as a bridge between software applications and the hardware of a computer. The kernel manages system resources, such as the CPU, memory, and devices, ensuring everything work together smoothly and efficiently.
- **Hardware:** Hardware refers to the physical components of a computer system or any electronic device. These are the tangible parts you can touch and see — such as the keyboard, mouse, monitor, CPU, hard drive, and even internal circuits. Hardware is typically directed by the software to execute any command or instructions.

Advantages of Windows

- **Hardware compatibility:** Almost every computer hardware manufacturing industry supports Microsoft Windows. This makes the users buy any random computer manufacturing brand and get the latest version of pre-loaded Microsoft Windows.
- **Ease of Use:** Microsoft Windows has developed by far the most user-friendly OS in the market, keeping in mind that it serves the purpose of most types of markets in the world. It's the most preferred OS for personal computers.

Disadvantages of Windows

- **Expensive:** Microsoft is a closed-source OS and the license cost is really high. It's not possible for every class of society to buy a new license every time one is expired.
- **Poor Security:** Windows is much more prone to viruses and malware in comparison to other OS like Linux or Mac
- **Not reliable:** Windows starts to lag with time and eventually needs booting every time and now to get back the initial speed.

Elements of the Windows Operating System

- **Desktop:** The desktop is the primary interface that users interact with after logging into Windows. It contains icons for files, folders, shortcuts.
- **Taskbar:** Located at the bottom of the screen, the taskbar shows open applications, system notifications, and the Start menu.
- **Start Menu:** A critical feature in Windows, it provides quick access to programs, settings, and power options. It is located at the bottom-left corner of the screen.
- **Windows Explorer:** This is the file management tool in Windows. It allows you to browse files, folders, and connected devices.
- **Task Manager:** A tool that provides information about processes, programs, and the system's overall health (e.g., CPU usage, memory usage). It can also be used to end tasks or restart applications.
- **File Explorer:** A specific part of the operating system that allows users to search for, open, and organize files. It is critical for file management.
- **Recycle Bin:** A temporary storage location for deleted files.

Uses of menu

Menus are a fundamental part of the Windows operating system (OS) user interface. They provide a way for users to interact with and control programs and settings in a structured and efficient manner.

Some of the most common types of menus used in the Windows OS:

- **Start Menu:** A central hub accessed by clicking the Start button or pressing the Windows key, it provides access to all applications, settings, files, and power options. You can pin frequently used items here for quick access.
- **Menu Bars:** Found at the top of most application windows, they provide a list of commands organized into pull-down menus (e.g., File, Edit, and View).
- **Context Menus (Pop-up or Shortcut Menus):** These menus appear when you right-click on an item, displaying a list of actions that are relevant to that specific object or area.
- **System Menu (Window Menu):** A pop-up menu that appears when you click the icon in the upper-left corner of a window or press Alt + Space. It allows you to perform actions like moving, resizing, or closing the window.
- **Quick Settings:** A modern menu, typically accessed from the taskbar icons for volume, network, or battery, to quickly adjust common system settings.

Tools

Windows OS includes built-in tools for maintenance, troubleshooting, and management, such as System Restore, Disk Cleanup, Resource Monitor, and Event Viewer. Other key tools include Remote Server Administration Tools (RSAT) for managing servers, and the Media Creation Tool for installing Windows.

Maintenance and troubleshooting

- **Disk Cleanup:** Frees up disk space by removing unnecessary files.
- **Resource Monitor:** Displays real-time data on system resource usage, such as CPU, memory, disk, and network activity.
- **Event Viewer:** Logs system events and errors, helping to diagnose problems.
- **System Restore:** Reverts the system to a previous state to fix recent system changes that caused issues.
- **Deployment Image Servicing and Management (DISM):** Used for servicing and preparing Windows images.

Management and administration

- **Remote Server Administration Tools (RSAT):** A collection of tools for managing roles and features on remote Windows servers.
- **Sysprep (System Preparation Tool):** Prepares a Windows installation for cloning and deployment on multiple computers.
- **User State Migration Toolkit (USMT):** Used to migrate user files, settings, and data from one PC to another during an OS deployment.
- **Control Panel:** Provides a centralized location to view and change system settings.

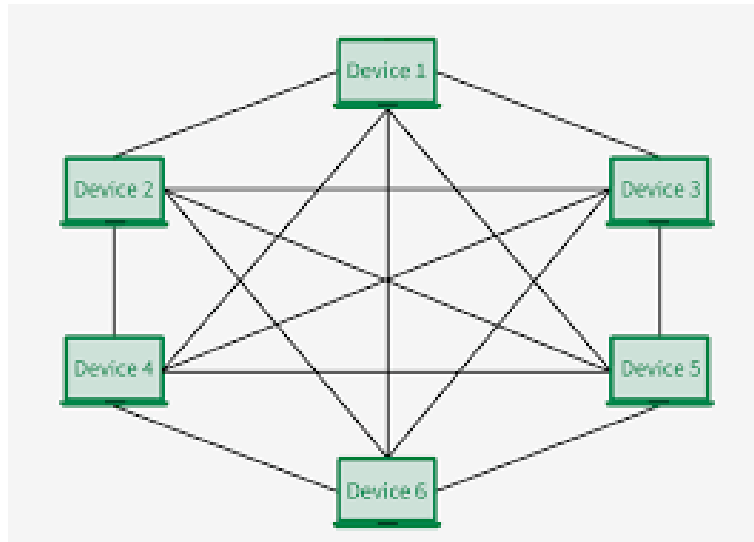
System and installation

- **Media Creation Tool:** Downloads the Windows installation files and creates a bootable USB drive for installing Windows on another PC.
- **Windows Terminal:** A modern terminal application that can host command-line shells like Command Prompt, PowerShell, and the Windows Subsystem for Linux.
- **Windows Subsystem for Linux:** Allows you to run a Linux environment directly on Windows.

COMPUTER NETWORKS

A computer network is a collection of computing devices that are connected with each other for the purpose of information and resource sharing among a wide variety of users.

A computer network is a collection of interconnected devices that share resources and information. These devices can include computers, servers, printers, and other hardware. Networks allow for the efficient exchange of data, enabling various applications such as email, file sharing, and internet browsing.

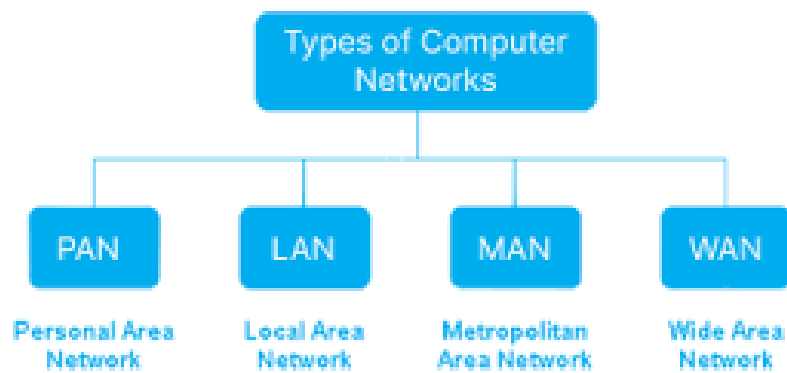


Uses of Computer Networks

- **Communication:** Email, chat, and video conferencing.
- **Resource Sharing:** Share printers, scanners, and files to save cost and effort.
- **Remote Access:** Access data and systems from anywhere.
- **Collaboration:** Work together on projects, share ideas, and review work.
- **E-commerce:** Enable online shopping and secure payments.
- **Education:** Support online learning, research, and student–teacher collaboration.

Types of Computer Networks

Computer networks are classified based on several factors, such as geographical area, ownership, architecture, topology, and transmission technology. The below is the classification of the networks:

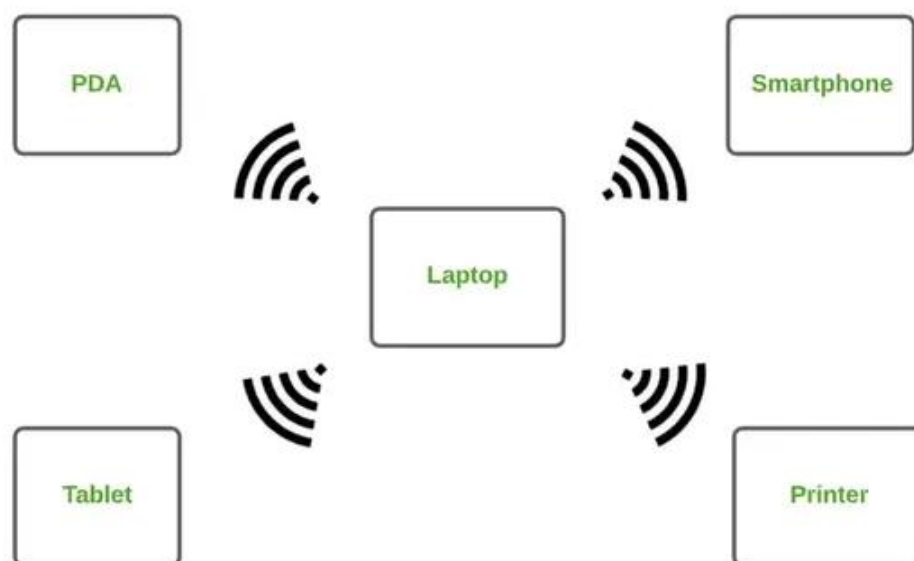


Personal Area Network (PAN)

PAN is the most basic type of computer network. It is a type of network designed to connect devices within a short range, typically around one person. It allows your personal devices, like smart phones, tablets, laptops, and wearable, to communicate and share data with each other. PAN offers a network range of 1 to 10 meters from person to device providing communication. Its transmission speed is very high with very easy maintenance and very low cost.

Examples:

Bluetooth connection between a phone and wireless ear buds, Infrared communication between TV and remote.

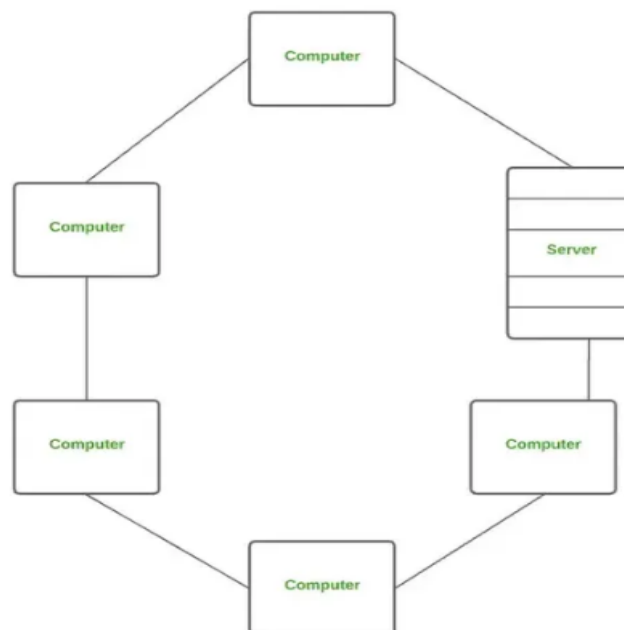


Local Area Network (LAN)

LAN is the most frequently used network. It is a computer network that connects computers through a common communication path, contained within a limited area, such as a single building or a floor within that building. A LAN encompasses two or more computers connected over a server.

The network equipment, like routers and switches, is typically managed by the owner of the network, for instance, you at home or the IT department at work. The two important technologies involved in this network are Ethernet and Wi-Fi. It ranges up to 2km & transmission speed is very high with easy maintenance and low cost.

Examples: Wi-Fi in a home or school, wired LAN in a company's office.



Metropolitan Area Network (MAN)

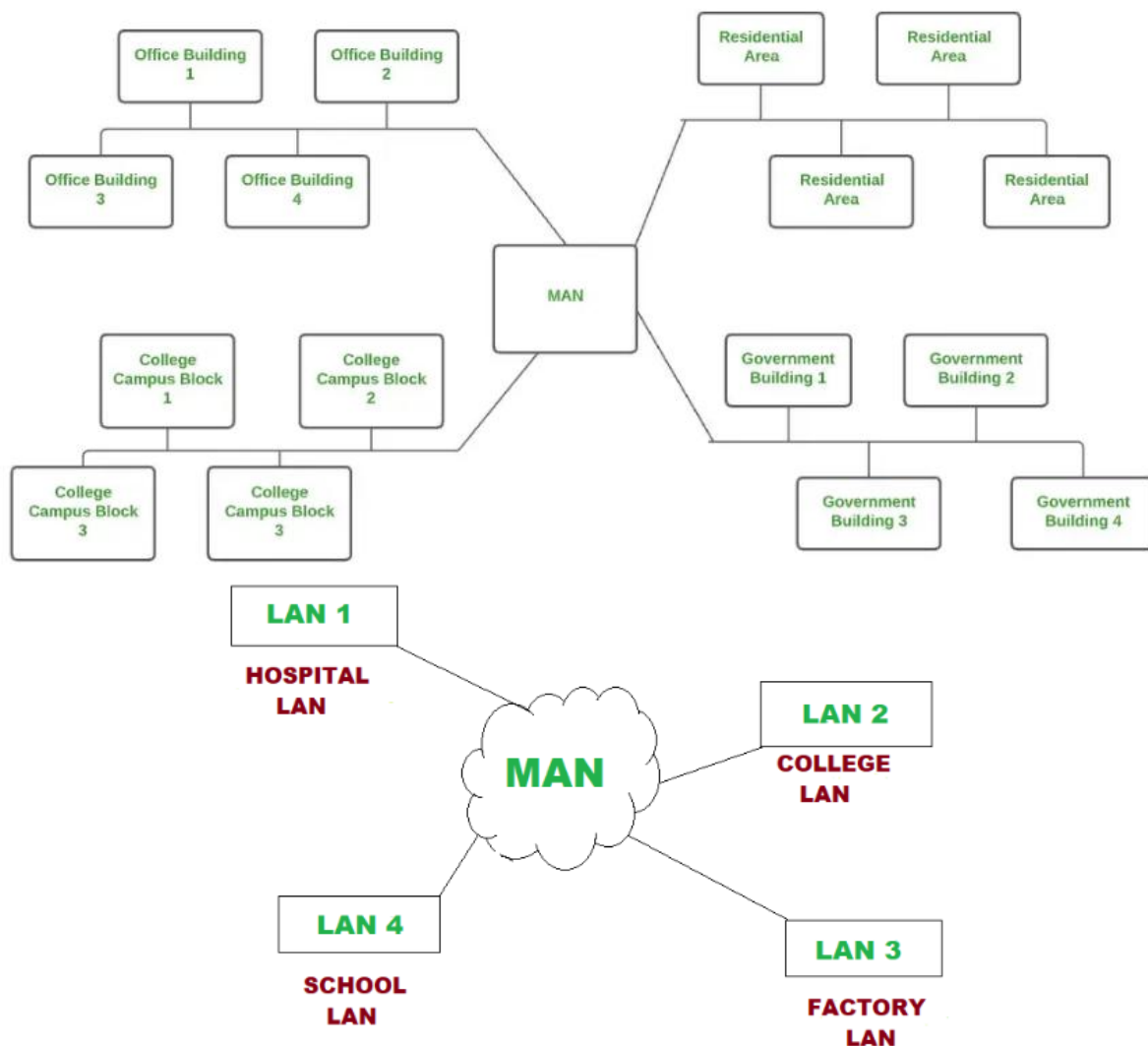
MAN stands for Metropolitan Area Network. It is a computer network that connects number of LANs to form larger network, so that the computer resources can be shared. This type of network covers larger area than a LAN but smaller than the area covered by a WAN which is designed to extend over the entire city.

MAN is specially designed to provide high-speed connectivity to the users in which the speed ranges in terms of Mbps. The architecture of MAN is quite complicated hence, it is hard to design and maintain.

Examples of MAN are networking in towns, cities, a single large city, a large area within multiple buildings, Cable TV network, used in hospitals to connect multiple buildings etc. It can cover the area which ranges from 5 to 50 km, which is capable of carrying from a group of buildings to the whole city.

In MAN, mostly used medium is optical fibers which results in high-speed connectivity. MAN network can use a variety of access technologies, such as wireless, fiber-optic, or copper-based connections, to provide connectivity to different devices and networks.

Hybrid topology: A MAN network may use a combination of different topologies, such as a ring, bus, or star topology, depending on the specific requirements of the network.



Advantages

- MAN offers high-speed connectivity in which the speed ranges from 10-100 Mbps.
- MAN can serve multiple users at a time with the same high-speed internet to all the users.

- MAN allows for centralized management and control of the network, making it easier to monitor and manage network resources and security.

Disadvantages

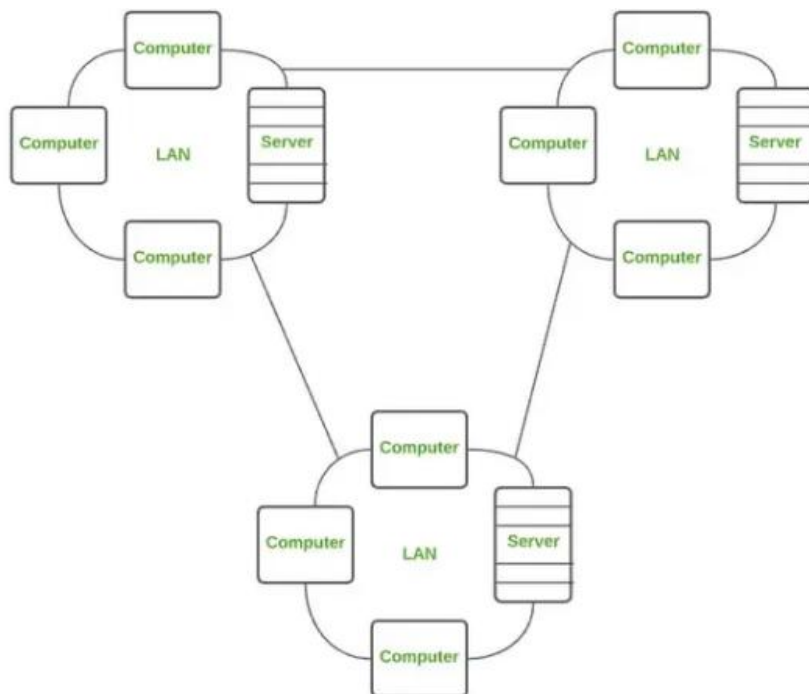
- The architecture of MAN is quite complicated hence, it is hard to design and maintain.
- This network is highly expensive because it required the high cost to set up fiber optics.

Wide Area Network (WAN)

WAN stands for Wide Area Network. It is a computer network that covers a large geographical area consisting of two or more LANs or MANs.

Example:

1. The internet, which connects networks across the globe.
2. Bank with branches and ATMs in different cities or countries, all linked together by a private WAN.
3. A university might use a WAN to connect its main campus to other campuses in different cities
4. Multinational corporations connect its headquarters and factories in different countries through a WAN to share production data and other information.



Advantages

- It covers large geographical area which enhances the reach of organization to transmit data quickly and cheaply.
- The data can be stored in centralized manner because of remote access to data provided by WAN.
- The travel charges that are needed to cover the geographical area of work can be minimized.
- WAN enables a user or organization to connect with the world very easily and allows to exchange data and do business at global level.

Disadvantages

- Traffic congestion in Wide Area Network is very high.
- The fault tolerance ability of WAN is very less.
- Noise and error are present in large amount due to multiple connection point.
- The data transfer rate is slow in comparison to LAN because of large distances and high number of connected system within the network.

TOPOLOGIES

A network topology is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them.

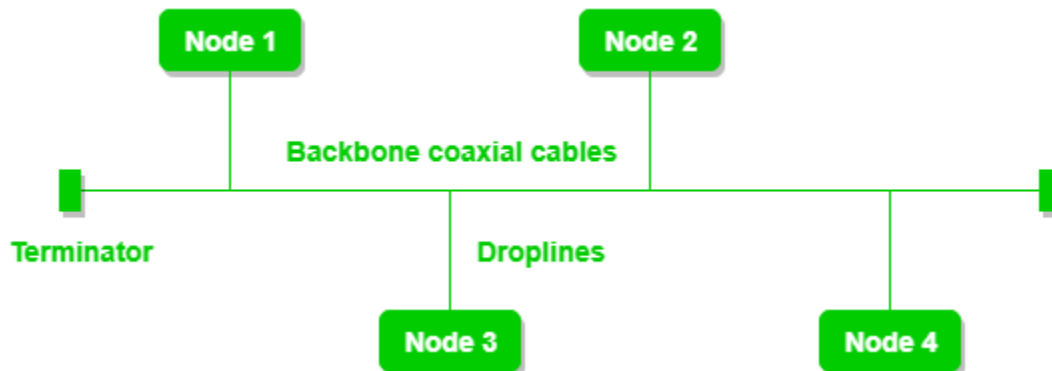
There are two main types of topology:

- **Physical Topology:** The actual physical layout of cables and devices.
- **Logical Topology:** How data moves across the network, regardless of physical layout.

Bus Topology

All of the devices in a bus topology network are linked together by a single cable, which is referred to as a "bus" and the cable is known as backbone cable. All of the network's devices can simultaneously receive the same signal due to the shared communication medium provided by this connection.

Bus topology carries transmitted data through the cable because data reaches each node; the node checks the destination address (MAC/IP address) to determine if it matches their address. If the address does not match with the node, the node does nothing more. But if the addresses of nodes match to addresses contained within the data, then they process knowledge.



Advantages

- It is the easiest network topology for linearly connecting peripherals or computers.
- It works very efficiently well when there is a small network.
- The length of cable required is less than a star topology.
- It is easy to connect or remove devices in this network without affecting any other device.
- Very cost-effective as compared to other network topology i.e. mesh and star
- It is easy to understand topology.
- Easy to expand by joining the two cables together.

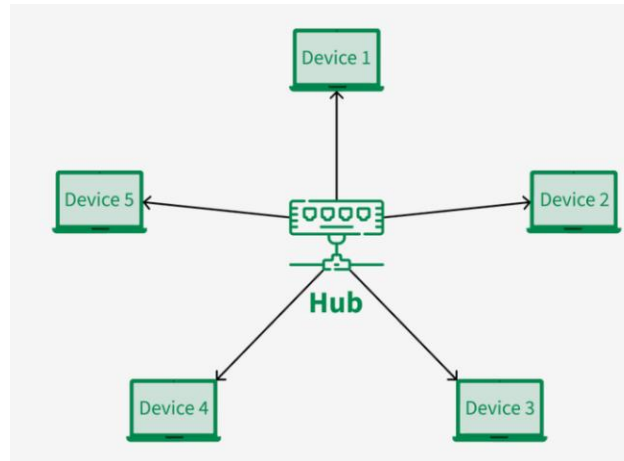
Disadvantages

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- Need terminators are required at both ends of the main cable.
- Additional devices slow the network down

Star Topology

In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub manages the data flow between the devices. If one device wants to send data to another device, it has to first send the information to the hub, and then the hub transmits that data to the required device.

The number of links required to connect nodes in the star topology is N where N is the number of nodes.



Advantages

- It is very reliable as if one cable or device fails then all the others will still work.
- It is high performing as no data collisions can occur.
- No disruptions to the network when connecting or removing devices
- Each device requires just one port i.e. to attach to the hub.
- If N devices are connected to each other in star, then the amount of cables required to attach them is N. So, it's easy to line up.

Disadvantages

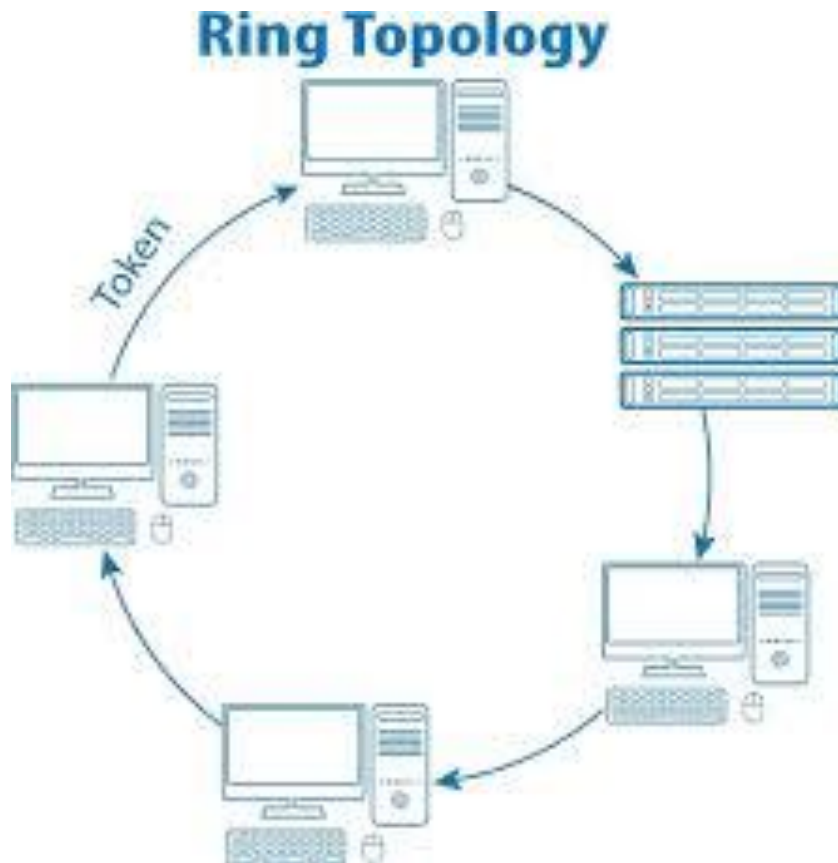
- If the hub goes down everything goes down, none of the devices can work without the hub.
- Hub requires more resources and regular maintenance because it's the central system of Star.
- Extra hardware is required (hubs or a switch) which adds to the cost.
- Performance is predicated on the one concentrator i.e. hub.
- More expensive than linear bus topology due to the value of the connecting devices (network switches).
- The cost of installation is high

Ring Topology

A ring topology is a network configuration where devices are connected in a circular, closed-loop pattern, with each device linked to exactly two neighbors. Data travels in a single direction, either clockwise or counter-clockwise, through each device until it reaches its destination.

If someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.

In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data where, Token passing is a network access method in which a token is passed from one node to another node & Token is a frame that circulates around the network.



One station is known as a monitor station which takes all the responsibility for performing the operations.

To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use. When no station is transmitting the data, then the token will circulate in the ring.

Advantages

- In this data flows in one direction which reduces the chance of packet collisions.
- There is no need of server to control the connectivity among the nodes in the topology.
- It is cheap to install and expand.
- Minimum collision.
- Speed to transfer the data is very high in this type of topology.
- The data transmission is high-speed.

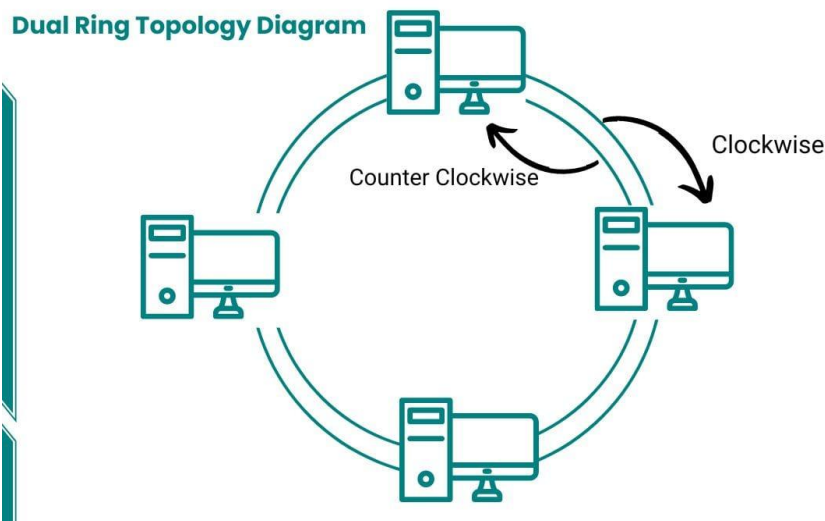
Disadvantages

- The failure of a single node in the network can cause the entire network to fail.
- Total dependence in on one cable.
- Addition and removal of any node during a network is difficult and may cause issue in network activity.
- Due to the Uni-directional Ring, a data packet (token) must have to pass through all the nodes.
- If one workstation shuts down, it affects whole network or if a node goes down entire network goes down.

Dual Ring Topology

A dual ring topology is a network configuration with two interconnected, concentric rings that improve reliability by providing a redundant data path.

Data can flow in opposite directions on the two rings, allowing traffic to be rerouted on the backup ring if the primary ring fails, thus minimizing downtime and increasing fault tolerance.



Advantages

- Provides a high level of network uptime because the failure of one ring does not stop the network.
- Creates a reliable network with a backup path for data in case of a link or node failure.

Disadvantages

- Implementing a dual-ring topology is more expensive than simpler configurations due to additional hardware requirements.

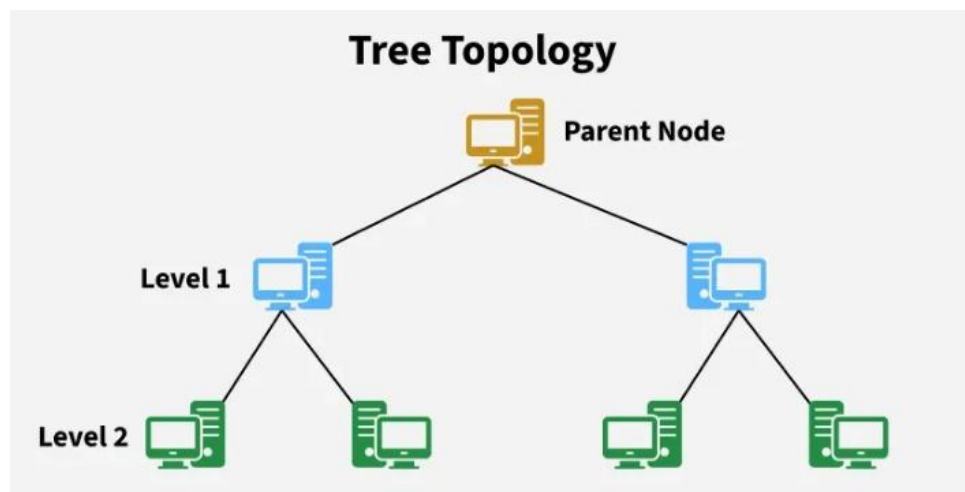
- Maintaining a dual-ring network can be more challenging, requiring specialized knowledge.

Tree Topology

Tree Topology has a hierarchical flow of data. Here, various secondary hubs are connected to the central hub which contains the repeater.

This data flow from top to bottom i.e. from the central hub to the secondary and then to the devices or from bottom to top i.e. devices to the secondary hub and then to the central hub.

Tree topology is a combination of bus and star topologies, resulting in a hierarchical structure.



Advantages

- It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.
- It allows the network to get isolated and also prioritize from different computers.
- We can add new devices to the existing network.

Disadvantages

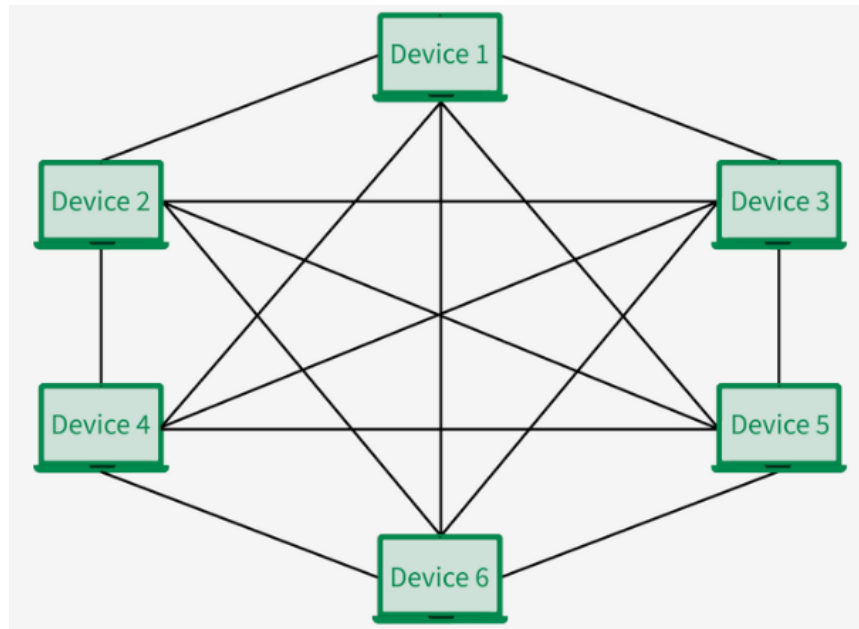
- If the central hub gets fails the entire system fails.
- The cost is high because of the cabling.
- If new devices are added, it becomes difficult to reconfigure.

Mesh Topology

Mesh Topology is a network configuration where every device is interconnected with every other device, providing multiple routes for data to travel. The nodes are connected to each other completely via a dedicated link during which information travels from nodes to nodes.

Suppose, the N numbers of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is $N-1$. If a mesh network has N nodes, then there are $N(N-1)/2$ links.

Each computer not only sends its signals but also transfer data from other computers. The connections within the mesh are often wired or wireless.



Advantages

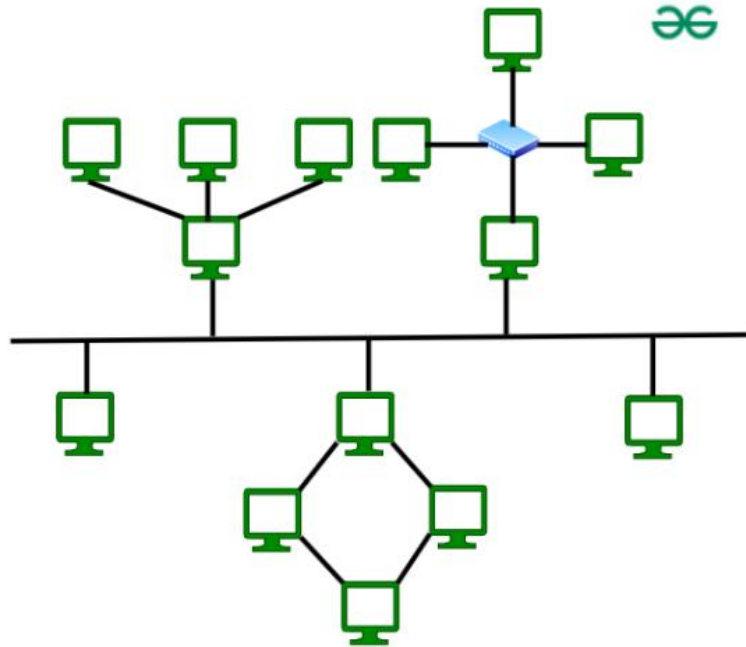
- In case of failure of a single device, the entire network didn't break.
- There is no traffic problem as there is a dedicated point to point links for every device.
- Mesh Topology provides high privacy and security.
- Data transmission is more consistent because failure doesn't disrupt its processes.
- Adding new devices won't disrupt transmission of data.

Disadvantage

- Mesh Topology is costly as compared to the other network topologies i.e. star, bus, point to point topology.
- Installation of nodes is difficult in mesh topology.
- Power requirement is higher as all the nodes will need to remain active all the time and share the load.
- More maintenance is required in mesh topology.

Hybrid topology

A hybrid topology is defined as a network topology that combines two or more different network topologies. A hybrid topology can be a combination of bus topology, ring topology and mesh topology. The selection of different types of network topologies combined together depends upon the number of computers, their location, and the required performance.



Advantages

- This topology is very flexible.
- The size of the network can be easily expanded by adding new devices.

Disadvantages

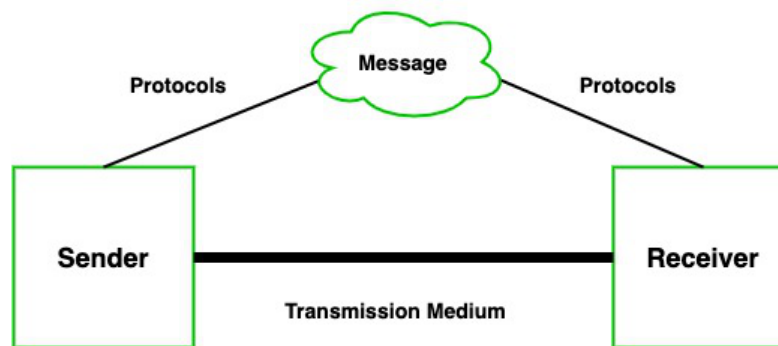
- It is challenging to design the architecture of the Hybrid Network.
- Hubs used in this topology are very expensive.
- The infrastructure cost is very high as a hybrid network requires a lot of cabling and network devices.

Network Topology is important because it defines how devices are connected and how they communicate in the network. Here are some points that define why network topology is important.

- **Network Performance:** Upon choosing the appropriate topology as per requirement, it helps in running the network easily and hence increases network performance.
- **Network Reliability:** Some topologies like Star, Mesh are reliable as if one connection fails, they provide an alternative for that connection, and hence it works as a backup.
- **Network Expansion:** Choosing correct topology helps in easier expansion of Network as it helps in adding more devices to the network without disrupting the actual network.
- **Network Security:** Network Topology helps in understanding how devices are connected and hence provides a better security to the network.

DATA COMMUNICATION

Data communication is the process of exchanging data between two or more devices over a transmission medium, such as cables or wireless signals.



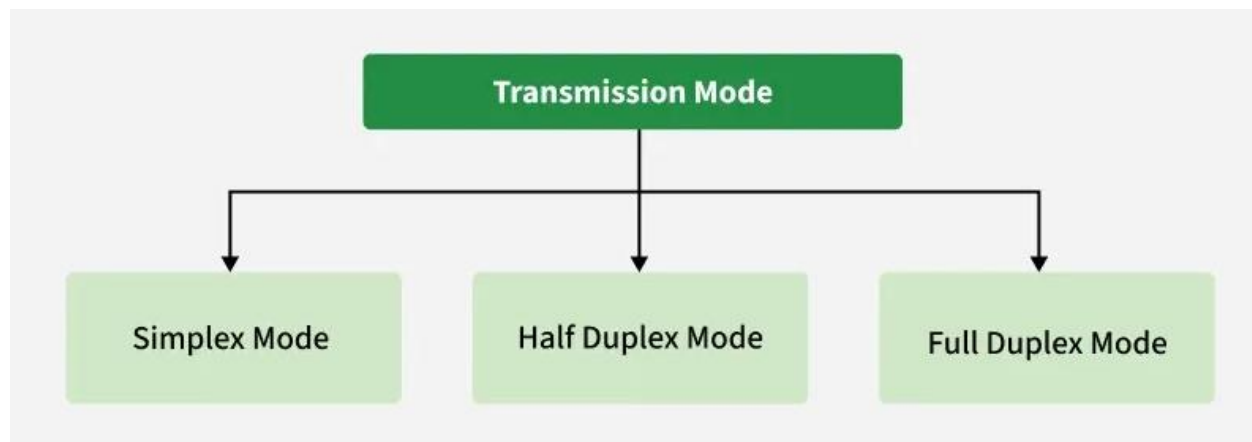
Components of Data Communication

- **Message:** A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.
- **Sender:** It is simply a device that sends data messages. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.
- **Receiver:** It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.
- **Transmission Medium / Communication Channels:** Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless media.
- **Protocol:** A protocol is a set of rules governing data communication, essential for devices to understand each other.
 1. **TCP (Transmission Control Protocol):** It is responsible for dividing messages into packets on the source computer and reassembling the received packet at the destination or recipient computer. It also makes sure that the packets have the information about the source of the message data, the destination of the message data, the sequence in which the message data should be re-assembled, and checks if the message has been sent correctly to the specific destination.
 2. **IP (Internet Protocol):** IP is responsible for handling the address of the destination computer so that each packet is sent to its proper destination.

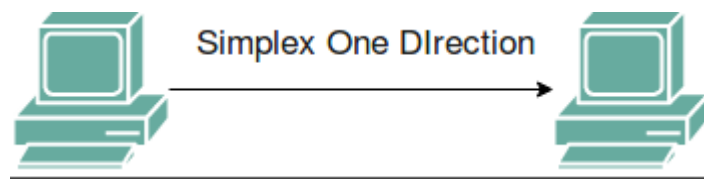
Type of data communication or Data flow modes

Transmission modes also known as communication modes, define how data is transferred between devices in a network or bus system.

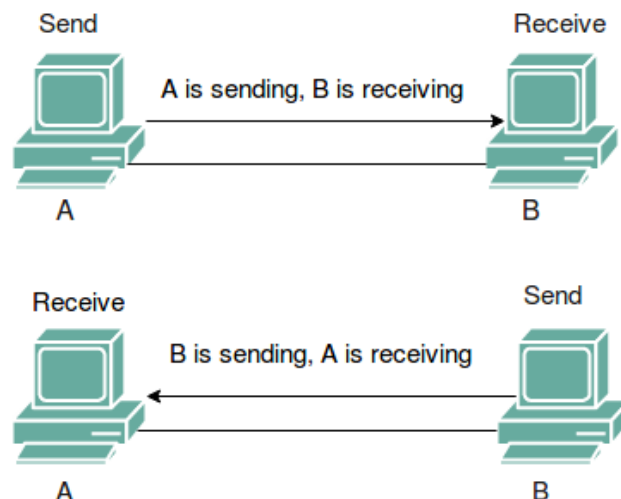
They are classified into three types



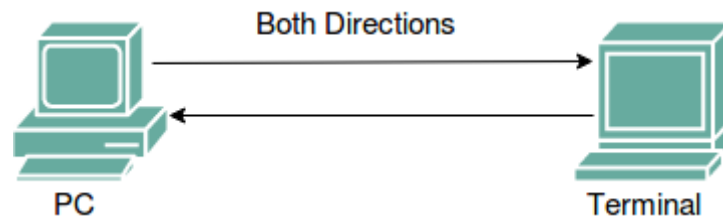
- **Simplex Communication:** It is one-way communication or we can say that unidirectional communication in which one device only receives and another device only sends data and devices use their entire capacity in transmission. For example, entering data using a keyboard, listening music using a speaker, etc.



- **Half Duplex communication:** It is a two-way communication, or we can say that it is a bidirectional communication in which both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. For example, walkie-talkie.

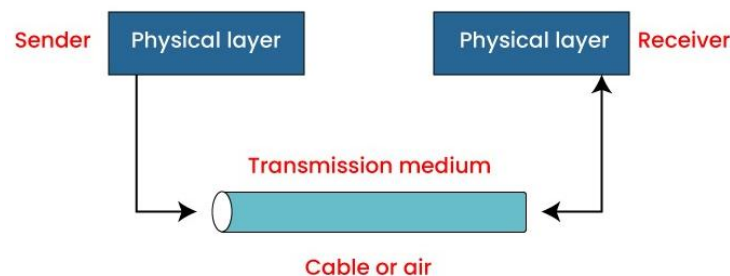


- **Full-duplex communication:** It is a two-way communication or we can say that it is a bidirectional communication in which both the devices can send and receive data at the same time. For example, mobile phones, landlines, etc.

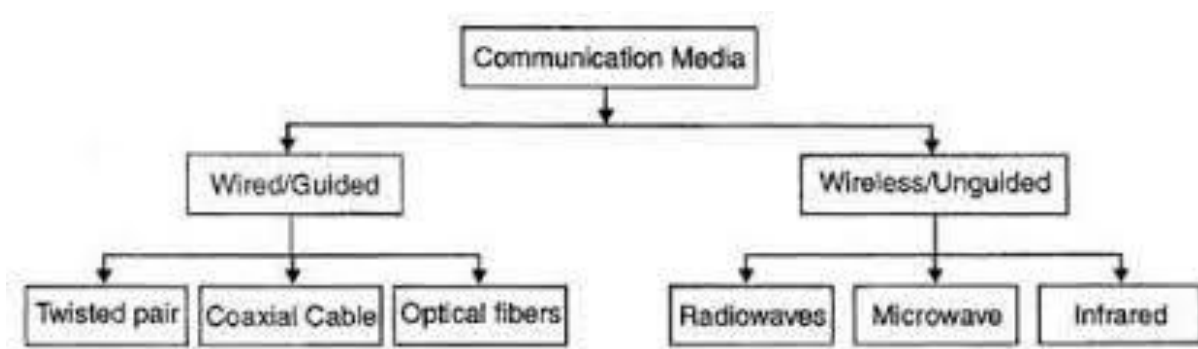


Communication Channels or Transmission Medium or Transmission Technology

Transmission Media is a method of establishing a communication medium to transmit and receive information in the form of electromagnetic signal waves. Workstations can be connected by either wired media or wireless media. It is also known as a transmission medium. The transmission medium or channel is a link that carries messages between two or more devices.



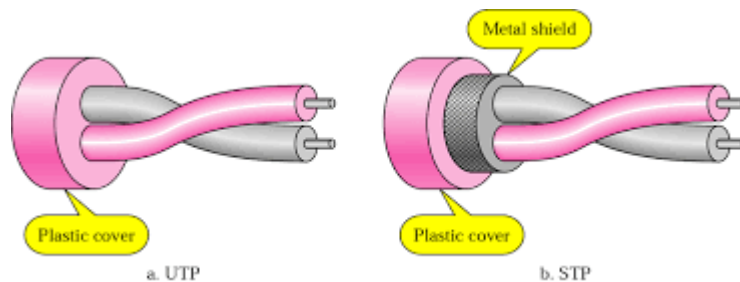
Communication media into two categories:



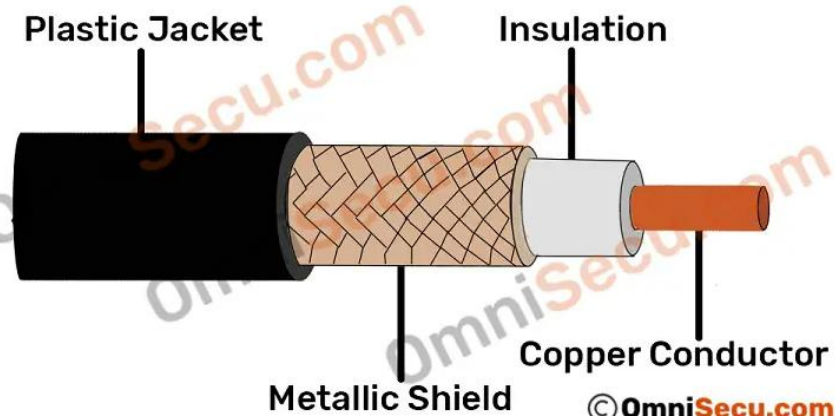
Guided media

There are physical connection exits between two ends source and destination.

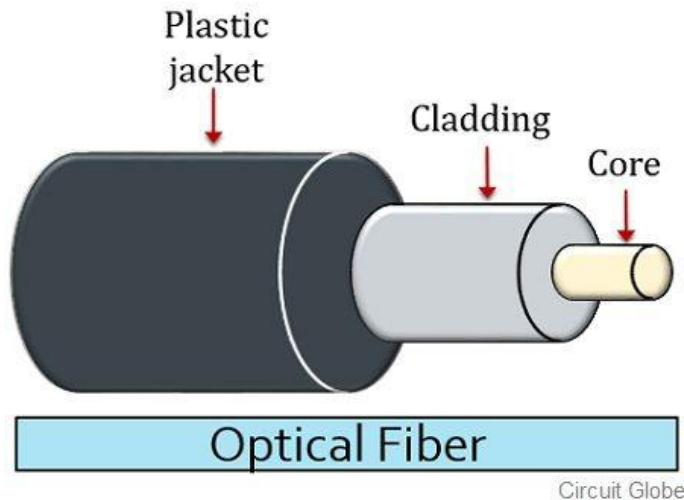
- **Twisted Pair Cable:** It is made up of insulated copper wires twisted together that are why it called is as twisted pair cable. Mostly used in telephone networks and Ethernet cables. It is affordable.
 - a. **Unshielded Twisted pair (UTP):** Unshielded means, the cable has the normal shielding and doesn't have any extra protection to secure from other interfaces.
 - b. **Shielded Twisted pair (STP):** As the name suggests that it has a shield to protect from other interfaces.



- **Coaxial Cable:** In coaxial cable one central conductor surrounded by insulation, a metallic shield, and an exterior cover. This type of cable used in television networks and for long-distance communication lines. It provides better protection against interruption than twisted pair cables.



- **Optical Fiber:** This cable uses light signals to transfer data. Made up of glass or plastic fibers, it provides very high bandwidth and low signal reduction, making it ideal for long-distance and high-speed data transmission. It is immune to electromagnetic obstruction but more expensive than copper cables.



Unguided Media

Unguided media means it is wireless communication between source and destination. The medium may be water, air, etc.

- **Microwave:** Used for long distances, antennas will be used placed on towers to send beams. It has a transmitter and receiver.



- **Radio wave:** It carries radio frequencies. Some transmitters and receivers have antennas to carry the signals.
- **Infrared:** Infrared is used for nearby communication, such as remote controls and short-range data transfer (e.g., between a laptop and a mobile device).

Difference between the Guided Media and Unguided Media

Guided Media	Unguided Media
The guided media is also called wired communication or bounded transmission media.	The unguided media is also called wireless communication or unbounded transmission media.
The signal energy propagates through wires in guided media.	The signal energy propagates through the air in unguided media.
Used to perform point-to-point communication	Unguided media is generally suited for radio broadcasting in all directions.
It is affordable.	It is costly.
It sends out a signal that indicates which way to go.	It does not indicate which way to travel.
For a shorter distance, this is the best option	For longer distances, this method is used.
By adding more wires, the transmission capacity can be increased in guided media.	It is not possible to obtain additional capacity in unguided media.